G1 locomotion on the updated Isaac Lab stack

RSL PPO, Warp PPO and Torch FlashSAC, compared at equal collected experience with common policy evaluations.

20 October 2026 · Three seeds per learner · Isolated NVIDIA L40 allocations · 50.38 million transitions per run · Download paper PDF

Abstract

At the common final budget, 9 of nine policies passed the forward-walking gate. Warp PPO's mean learning-phase cost was 26.1% lower than stock RSL PPO; its recorded training-loop cost was 4.5% lower. Including the matched smoke and evaluation protocol, its allocation wall cost was 7.1% higher. The tables and curves report policy quality, collected experience and elapsed time separately. These three seeds describe this task and stack; they do not establish a universal backend advantage.

9/9

Final policies passing the common forward-walking gate. A pass combines survival, velocity tracking and commanded forward movement.

26.1% lower

Warp PPO learning-phase cost relative to stock RSL PPO, using three-seed means. This compares the matched PPO recipes.

4.5% lower

Warp PPO recorded training-loop cost relative to stock RSL PPO. All 2,050 rows include collection and learning; the protocol table also includes smoke and evaluations.

> **Three different questions.** Learning curves measure experience needed to obtain a useful policy; runtime measures the cost of the matched training budget; deterministic evaluations measure behavior. A faster update does not establish better learning quality.

01 How much experience produced useful behavior?

First observed passing iterations for seeds 0 / 1 / 2: RSL PPO: 1000 / 1500 / 1500; Warp PPO: 1000 / 1000 / 1000; FlashSAC: 500 / 500 / 500. Initial policies and five trained checkpoints are scored. Passing checkpoints are sparse observations, and a policy can later regress.

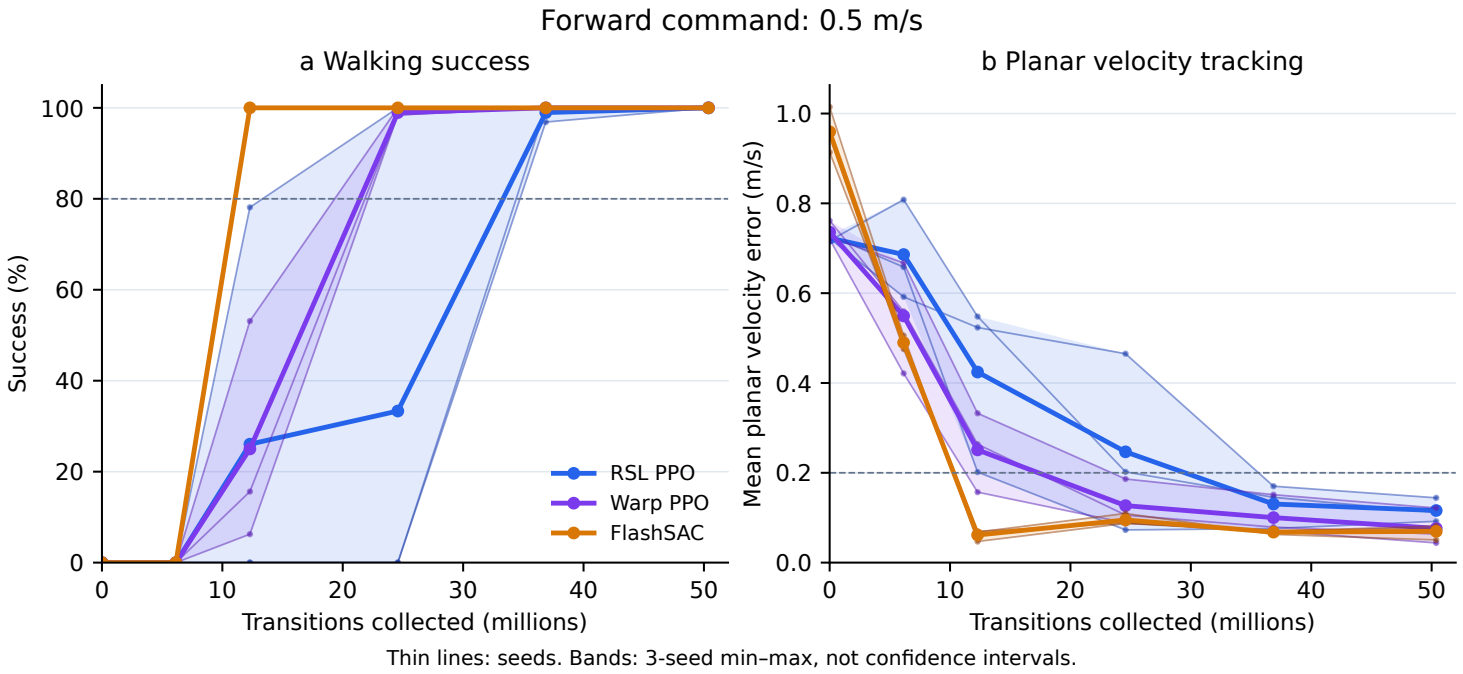


Figure 1. Checkpoints include the fresh initial policy. Thin lines show seeds; thick lines show their mean; bands show the three-seed minimum–maximum range, not confidence intervals. Lines connect observations without estimating intermediate behavior.

> **Reading the figure.** Every point evaluates 64 worlds. Thin lines are seeds; thick lines are means; bands are their observed minimum–maximum range. Bands are **not confidence intervals**. Native commands include standing and changes in direction; native tracking success and fixed-forward walking success are different metrics.

▼ **Individual passing checkpoints and regressions**

The sample interval runs from the preceding observed checkpoint to the first pass. It does not locate an exact convergence time. A later failed checkpoint is retained.

Learner	Seed	First observed pass (iteration)	Observed sample interval (M)	Later failing checkpoints
RSL PPO	0	1000	(12.288, 24.576]	None observed
RSL PPO	1	1500	(24.576, 36.864]	None observed
RSL PPO	2	1500	(24.576, 36.864]	None observed
Warp PPO	0	1000	(12.288, 24.576]	None observed
Warp PPO	1	1000	(12.288, 24.576]	None observed
Warp PPO	2	1000	(12.288, 24.576]	None observed
FlashSAC	0	500	(6.144, 12.288]	None observed
FlashSAC	1	500	(6.144, 12.288]	None observed
FlashSAC	2	500	(6.144, 12.288]	None observed

02 Where is the training time spent?

The left panel separates collection and learning in the recorded production loop: the sum of all 2,050 iteration rows, including early iterations. The right panel includes the rest of the production process, such as explicitly separated preparation, checkpoint writes and logging. Neither panel includes preceding smoke or subsequent common evaluation commands.

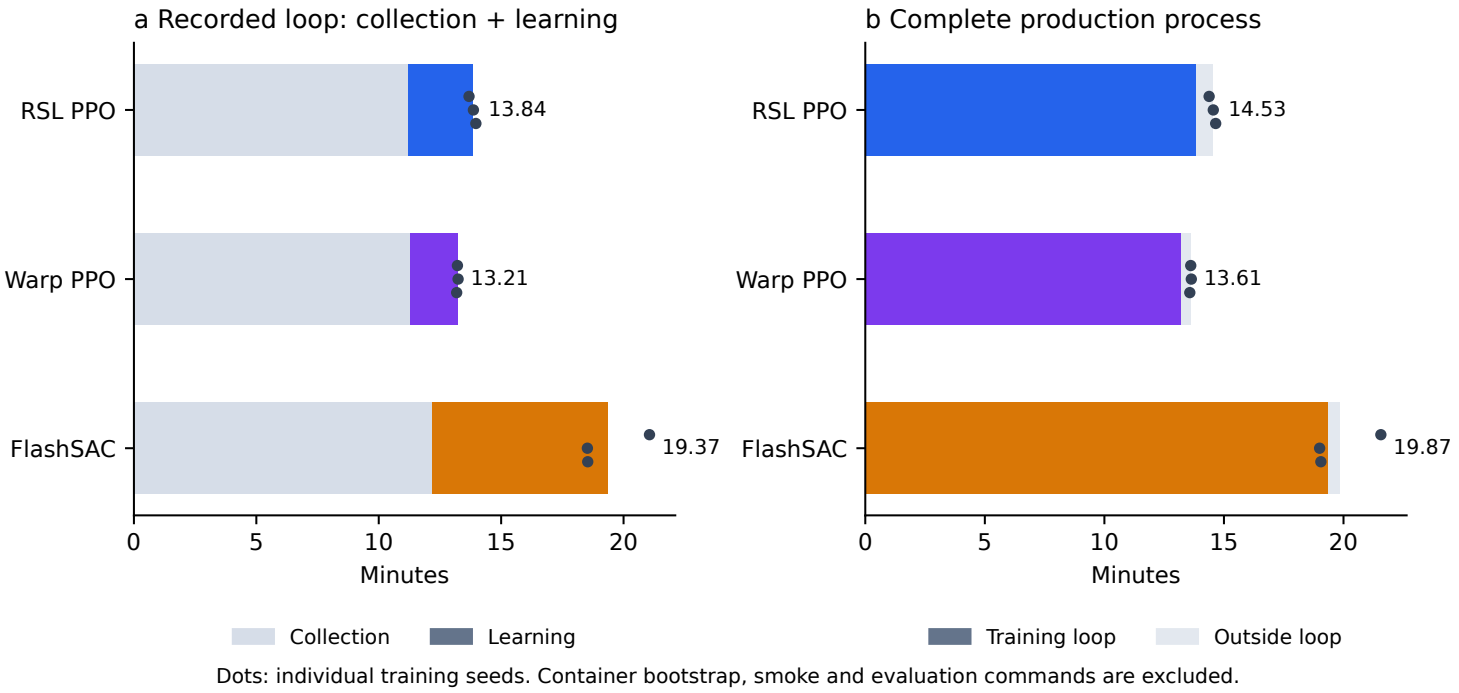


Figure 2. Training-loop attribution and complete production-process cost. Bars show three-seed means; dots show individual seeds. Outside-loop cost includes preparation, logging and checkpoint writes. Smoke, offline evaluations, container bootstrap, queueing and retrieval are excluded.

Learner	Recorded loop (min) ↓	Collection (min)	Learning (min)	Production process (min) ↓	Steady transitions/s ↑
RSL PPO	13.84	11.21	2.63	14.53	60,863
Warp PPO	13.21	11.27	1.94	13.61	63,717
FlashSAC	19.37	12.20	7.17	19.87	45,536

Three-seed means. Each algorithm used a separate dedicated L40 allocation, with three seeds run serially within it. GPU clocks were not locked, and the algorithms were not rerun in rotation on the same physical GPU. Small differences in steady throughput can reflect GPU or node variation; they are descriptive measurements, not statistical proof of a backend advantage.

The steady-throughput column excludes exactly the first five iteration rows; this does not guarantee all later compilation has finished. The recorded loop total includes all rows. Explicit learner warmup is measured separately; RSL records zero such warmup, while initial physics graph capture can occur inside its first rollout. Compilation caches are reused within an algorithm's allocation.

> **Phase timing conventions.** PPO uses synchronized wall timing; FlashSAC interleaves replay updates with collection and attributes the phases using CUDA events. Their splits do not represent identical gradient work. Production-process costs exclude smoke, queueing, container setup and artifact retrieval.

Complete matched protocol in each successful allocation

Learner	Protocol wall (min) ↓	3 smoke trains (min)	3 production trains (min)	3 smoke evaluations (min)	3 common evaluations (min)
RSL PPO	53.46	3.48	43.85	1.32	4.80
Warp PPO	57.25	9.67	41.09	1.33	5.15
FlashSAC	78.06	10.85	59.88	1.86	5.47

Each row spans the first seed-0 smoke-training command start through the final seed-2 common-evaluation command end. It includes all three smoke-training processes, all three production-training processes, and all smoke/common evaluation commands. The wall span also includes small gaps between commands. Values come from the successful allocation's command receipts, whose hashes are retained in public provenance.

This is successful-allocation protocol wall time, not cold learning alone. It exposes first-use compilation and preparation performed during smoke instead of silently dropping those costs from the comparison. It excludes operating-system/dependency installation, failed earlier workflow attempts, audit/render steps and optional component probes. Caches are reused within each allocation; the table does not measure independently cache-cleared machine starts.

03 What do the final policies do?

The final checkpoint has the same collected-transition budget for every learner. Velocity error measures commanded movement; survival records completion of a native 20-second episode. Surviving while standing does not suffice for the forward-walking criterion.

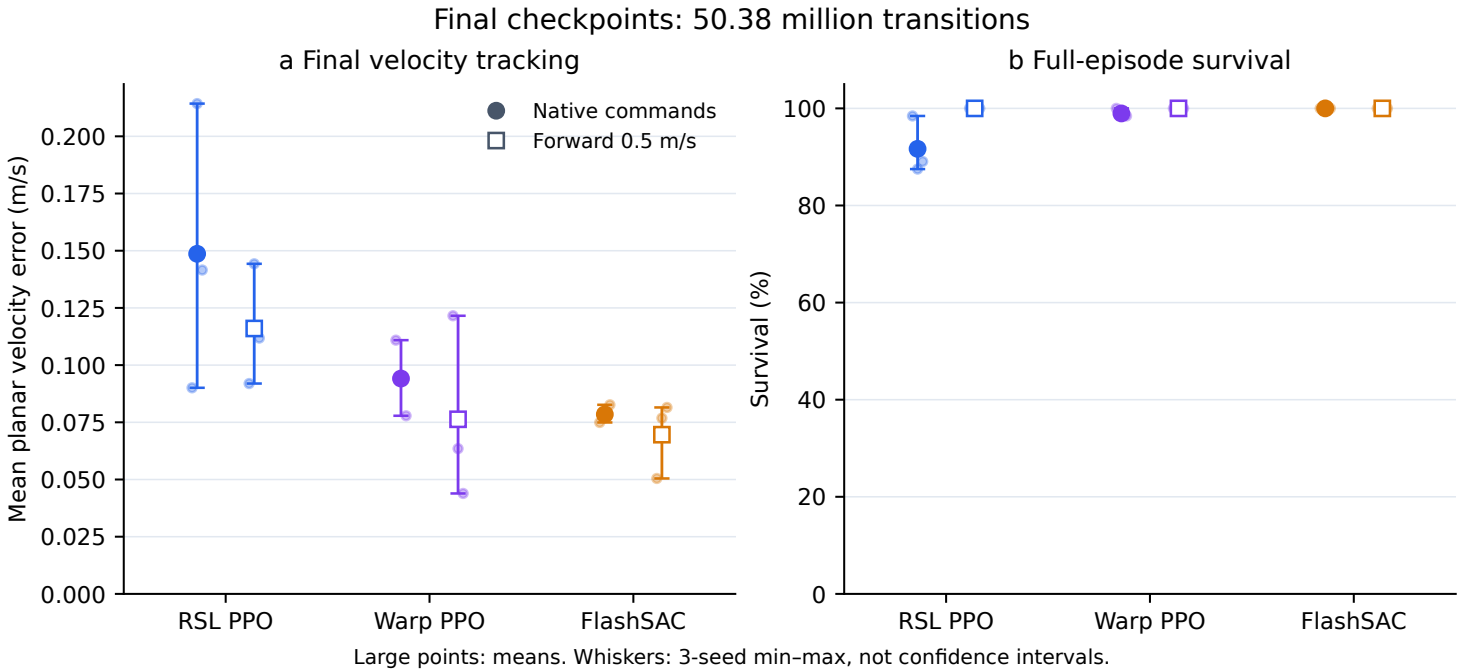


Figure 3. Final checkpoints at 50.38M collected transitions. Solid circles: native commands; open squares: fixed forward commands. Individual seed points accompany mean and min–max markers. Survival alone does not establish successful walking.

Fixed forward command, 0.5 m/s

Learner	Planar error (m/s) ↓	Survival ↑	Walking success ↑
RSL PPO	0.116 0.092–0.144	100.0%	100.0%
Warp PPO	0.076 0.044–0.122	100.0%	100.0%
FlashSAC	0.070 0.050–0.081	100.0%	100.0%

Native command distribution

Learner	Planar error (m/s) ↓	Survival ↑	Native tracking ↑
RSL PPO	0.149 0.090–0.214	91.7%	91.7%
Warp PPO	0.094 0.078–0.111	99.0%	99.0%
FlashSAC	0.078 0.075–0.083	100.0%	100.0%

The smaller numbers show the three-seed error range. Survival and success average the three seeds, each evaluated over 64 worlds. These metrics do not certify natural gait, balanced foot support or suitability for hardware.

04 What changed since the previous baseline?

This refresh starts from upstream Isaac Lab main `159af106ce`, with a small RoboLearn integration layered on top. The current registered G1 task settings and stock PPO recipe are retained. The physics and training dependency versions are recorded from the actual experiment environment.

Dependency	9 October cohort	20 October cohort
torch	2.12.0+cu130	2.11.0+cu130
rsl-rl-lib	5.5.1	5.5.1
warp-lang	1.17.0	1.17.0
warp-nn	0.4.0	0.4.1
newton	1.6.1	1.8.0.dev0
mujoco-warp	3.12.0	3.15.0
robolearn-rl	0.1.0	0.3.0

Historical context at the same collection budget

Learner	Old loop (min)	New loop (min)	Loop cost change	Old error (m/s)	New error (m/s)
RSL PPO	13.93	13.84	0.6% lower	0.114	0.116
Warp PPO	13.59	13.21	2.8% lower	0.090	0.076
FlashSAC	20.51	19.37	5.6% lower	0.053	0.070

> **A stack refresh, with several moving parts.** New versus old results include changes to Newton, MuJoCo Warp, Torch and Warp-NN, along with the maintained RoboLearn adapter. They do not isolate a learner or kernel change. The old cohort had 90 trained-policy evaluations; this cohort has 108 because it also scores each initial policy. Original results and raw JSON remain unchanged in the 9 October report.

Warp-NN 0.4.1 changes its default two-dimensional tiles from 16×16 to 16×64 and changes some kernel launch settings. That changes actual forward matrix work. It does not itself establish that the current FP32 Warp learner matches Torch throughput.

05 Should FlashSAC move entirely to Warp?

The primary nine-run cohort trains **Torch FlashSAC**; it does not train a Warp FlashSAC walking policy. Optional measurements below isolate the experimental Warp learner with synthetic replay and G1-sized observations/actions, after all three primary Warp PPO seeds finish.

Torch FP32 eager vs Warp FP32: the captured Warp update took 2.59× the Torch update time. Captured Warp update cost was 1.5% lower than its eager mode in this single-process probe.

Torch AMP compiled vs Warp FP32: the captured Warp update took 6.86× the Torch update time. Captured Warp update cost was 1.5% lower than its eager mode in this single-process probe.

Comparison	Backend	Update (ms/call) ↓	Actor (ms/call) ↓	Actor mode	Ordered startup (s)	Actual C/A/T updates
Torch FP32 eager vs Warp FP32	Torch	11.527	0.644	eager	3.95	100/50/50
Torch FP32 eager vs Warp FP32	Warp eager	30.312	0.820	eager	10.45	100/50/50
Torch FP32 eager vs Warp FP32	Warp captured	29.869	0.824	eager	1.39	100/50/50
Torch AMP compiled vs Warp FP32	Torch	4.353	0.168	compiled	494.10	100/50/50
Torch AMP compiled vs Warp FP32	Warp eager	30.312	0.848	eager	1.59	100/50/50
Torch AMP compiled vs Warp FP32	Warp captured	29.868	0.834	eager	1.46	100/50/50

Synthetic replay and G1-shaped networks; no physics, MDP or walking evaluation. The default actor/critic widths are 128/256 with two blocks each, observation/action widths 123/37, N=1,024 and update batch=2,048. Replay capacity is 65,536; sixteen vector batches ingest 16,384 transitions. Twenty warmup calls precede 100 update calls (100 critic attempts and 50 actor/temperature attempts). AMP skips are reported separately. Actor timings are FP32; Warp actor inference is eager even when its learner is captured. The FP32 label describes the configured recipe; effective TF32/IEEE math modes and actual tensor instructions were not recorded, so this is not a strict IEEE-parity measurement. These measurements are descriptive, without seed uncertainty bars.

Ordered preparation, not independent cold starts. Both probes run after the primary Warp PPO seeds and share that allocation's compilation caches. Within each process the order is Torch, Warp eager, then Warp captured; later stages reuse previously compiled modules. The AMP compiled probe follows the FP32 eager probe and its Warp stages can reuse the earlier process's cache. Startup includes construction, common-weight loading, replay ingestion, preparation and warmup, but these startup values do not measure independent cold starts.

How to interpret the decision

Use whole-learner update timing, startup cost and actual optimizer work together. If captured Warp remains slower, more capture alone is unlikely to fix matrix arithmetic or gradient reductions. A hybrid implementation can keep Warp storage, optimizer and surrounding kernels while using optimized matrix multiplication. A fully Warp implementation remains an engineering option; its speed needs to be demonstrated on the intended shapes and precision.

No direct policy-inference latency claim follows from training-loop timing. The synthetic actor measurements, when supplied, are a separate inference measurement with their own execution modes. Neither a component probe nor the earlier dense-kernel diagnosis establishes a universal speed limit for Warp.

06 Experimental method

Shared environment

`Isaac-Velocity-Flat-G1`; Newton MJWarp physics, Torch frontend and manager MDP. Rewards, observations, commands, PD action scale, resets and terminations use the current registered task.

Common collection budget

1,024 environments × 24 steps × 2,050 iterations = 50,380,800 transitions per run. Fresh policy/optimizer for seeds 0, 1 and 2. Initial episode counters are randomized consistently.

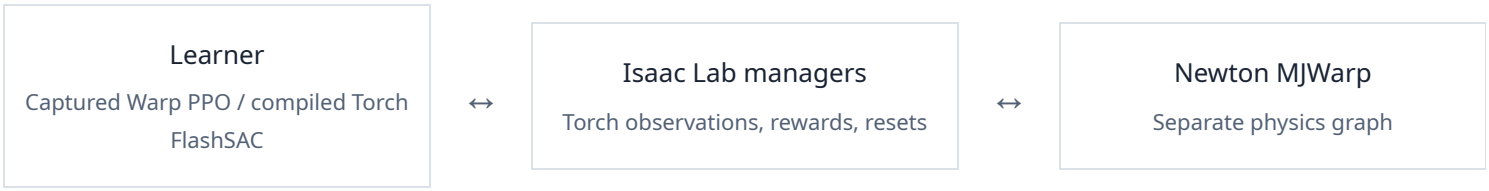
Common evaluation

64 worlds, maximum native 20-second episodes; deterministic actions, no observation corruption or pushes. Evaluation seed = 10000 + learner seed. Checkpoints 0, 250, 500, 1000, 1500 and 2050 under two command scenarios.

Walking gate

At least 80% walking success, mean planar error ≤ 0.2 m/s, and survival ≥ 90%. A walking world survives, tracks planar velocity within 0.25 m/s and yaw within 0.4 rad/s, and moves forward faster than 0.25 m/s.

Capture scope



Hardware assignment: one dedicated L40 allocation per algorithm; seeds 0, 1 and 2 run serially inside that allocation. The experiment did not lock GPU clocks or rotate algorithms on the same physical GPU.

This is not a unified physics–MDP–learning graph and does not train through physics gradients.

▼ Matched PPO and algorithm-specific FlashSAC work

Warp PPO matches the current stock G1 PPO recipe: ELU networks, unnormalized observations, unclipped Gaussian actions, scalar standard deviations, adaptive learning rate, five epochs and four minibatches. The saved configuration is checked against the registered recipe; optimized Warp Linear gradients are enabled.

FlashSAC retains the authors' replay, distributional-critic and bounded-action recipe, with compilation and AMP. It uses larger networks and different numbers/sizes of replay updates. Equal collected experience does not mean equal optimizer work or policy support. Saved optimizer counts distinguish attempted updates from actual AMP steps.

For seeds 0 / 1 / 2, FlashSAC attempted 49,101 actor updates per run and completed 49,039 / 49,037 / 49,039, with 62 / 64 / 62 AMP skips. Each run completed 98,202 critic and 49,101 temperature updates. Each PPO run completed 41,000 actor and critic updates.

Actual observation/action dimensions: `{"action_dim": 37, "critic_observation_dim": 123, "observation_dim": 123}`.

Interpretation and limits

- Three seeds provide descriptive evidence for this G1 task, hardware and software stack. They do not establish statistical significance or universal superiority.
- Evaluation checkpoints are sparse and retrospective. The first observed passing checkpoint can be followed by failure, and behavior between checkpoints is unknown.
- The setup is an Isaac Lab 3/Newton integration benchmark, not an exact replication of the FlashSAC paper's Isaac Lab 2.1/PhysX experiments or its real-robot deployment.
- Different algorithms and execution modes consume different gradient work, precision and preparation time. Matching numeric seed labels does not match every random draw.

07 Evidence and reproducibility

The report requires a complete passing cohort summary and verifies every raw comparison JSON hash before rendering. The finite-weight, budget and evaluation checks are retained in the evidence. Original raw files and all native checkpoints remain in their retrieved artifact bundles; public downloads contain an explicitly derived subset.

Download derived measurements JSON · Download all 108 evaluations CSV · Report provenance · Report file checksums

Learner	Isaac Lab integration	RoboLearn
RSL PPO	`0879f3bc7905932770f34f779cbaf6fb72eef0d5`	`7e872dbda8a0385a7fae227dbc6e61c608aa1696`
Warp PPO	`afcabbddb03bdd665bddd7c16ab931dd644cbe34`	`7e872dbda8a0385a7fae227dbc6e61c608aa1696`
FlashSAC	`afcabbddb03bdd665bddd7c16ab931dd644cbe34`	`7e872dbda8a0385a7fae227dbc6e61c608aa1696`

Source revisions are preserved per run. The table lists every integration and learner revision used by each algorithm; the public measurements retain the exact learner/seed mapping. Native RSL PPO retains its earlier valid integration snapshot. Warp PPO and FlashSAC use a follow-up registration-only configclass compatibility fix: construct G1FlatFlashSACRunnerCfg() before reading its algorithm_cfg, rather than accessing that field on the class. Native RSL never imports this registration module. The fix changes configuration construction before training, with the shared MDP, learner recipes and runtime dependencies checked separately. The cohort does not use one identical Isaac Lab integration commit.

▼ **Immutable raw measurement hashes**

Resolved MDP SHA256: `1b209baf98c7d5ff91c63c55ad06ab6a43ef2f9b6c72eca2ad787b3678124455`
Derived cohort summary SHA256: `4d80d065ff4327f8718031bff6fa95f8388cf38c90bd12a12447ff7bca6d78cc`

Learner	Seed	Raw comparison JSON SHA256
FlashSAC	0	`02327b8e830cf02d54f3d38da63a919a8d5dfe6467b1abe41920534cac73d72b`
FlashSAC	1	`b1d50435d605292f4f0864b9717475edf33ea0b412e5edf2c006ccc71b1f2f78`
FlashSAC	2	`3d2a684f4c198f3d1723aa6b7581341649f2cc2708c8ff8c633f2665f35a6ef6`
RSL PPO	0	`25cd976f0c8f897973320a3ff9d4a310dba698c02db4fc7892c963f0cfde990c`
RSL PPO	1	`a6689d25b2bde3ed6beac30e2a51aae49ac3ed894ffbdec49574d6009e824f7e`
RSL PPO	2	`dca5a2b2d6e434f6bfbd7d6923c2e58efd3cbb55033a13b0d0820633d5b5e390`
Warp PPO	0	`ccb414942af003b52a3826b210d43f508990bed2241149e70f82612efc0b31a4`
Warp PPO	1	`d2312207f1127d6c2f0c95f66794b76fc83549f9fc3560222f6018bbd46ce448`
Warp PPO	2	`136a1457d01c0cfceff12a64f0c9deac470ce2bd4c0d3523954ae0d99e27744c`

References and attribution

1. Donghu Kim, Youngdo Lee and coauthors. FlashSAC, 2026. Adapted official implementation. Original copyright and MIT license are retained.
2. John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford and Oleg Klimov. Proximal Policy Optimization Algorithms, 2017.
3. RSL-RL, Isaac Lab, Newton and MuJoCo Warp: their respective authors and contributors.
4. NVIDIA and contributors. Warp-NN and Warp. RoboLearn's Warp PPO independently implements the published PPO objective.
5. RoboLearn: full attribution and citations.

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